

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)

Assessment Tool Weightage: 10%

Code of Conduct:

I M. Pranay (192525382)_certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

M. Pranay

Signature of the student:

Rubrics for Evaluation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks (100)
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	

<i>Accuracy</i>	<i>10%</i>	<i>Completely accurate and covers all required points</i>	<i>Mostly accurate with minor omissions</i>	<i>Partially correct with missing points</i>	<i>Incorrect answers with major gaps</i>	
<i>Clarity</i>	<i>5%</i>	<i>Clear, neat, professional presentation</i>	<i>Understandable with minor issues</i>	<i>Limited clarity, readability issues</i>	<i>Poorly written, difficult to understand</i>	

1. An image contains high-frequency noise. Explain and justify the appropriate frequency domain filter, including its working and impact on image quality.

A.

Introduction

High-frequency noise appears as random variations in pixel intensity and reduces image quality. In the frequency domain, this noise is mainly present in the high-frequency components. To remove such noise, a Low-Pass Filter (LPF) is used because it allows low-frequency components to pass while suppressing high-frequency components.

Suitable Filter

Ideal Low-Pass Filter (ILPF), Butterworth Low-Pass Filter (BLPF), or Gaussian Low-Pass Filter (GLPF). Among these, the Gaussian Low-Pass Filter (GLPF) is preferred because it produces smooth results without ringing artifacts.

Working

1. Convert the image from the spatial domain to the frequency domain using the Fourier Transform (FFT).
2. Apply the Low-Pass Filter to retain low-frequency components and remove high-frequency noise.
3. Multiply the frequency spectrum with the filter.
4. Perform the Inverse Fourier Transform (IFFT) to obtain the filtered image.

Impact on Image Quality

- Reduces high-frequency noise effectively.
- Produces a smoother image.
- Preserves important low-frequency information.
- Excessive filtering may blur edges and fine details.

Advantages

- Efficient noise removal.
- Improves visual quality.
- Easy to implement.
- Suitable for preprocessing in image analysis.

Applications

- Medical image enhancement.
- Satellite image processing.
- Digital photography.
- Computer vision systems.

Conclusion

A Gaussian Low-Pass Filter is the most suitable choice because it effectively removes high-frequency

noise while preserving overall image quality.

2. You need to preserve low-frequency components while removing noise. Describe and apply a suitable filtering technique, explaining how it achieves the desired effect.

A. Introduction

Low-frequency components represent the major structures and smooth regions of an image. To preserve these components while removing unwanted noise, a Low-Pass Filtering technique is applied.

Suitable Technique

Butterworth Low-Pass Filter (BLPF).

Working

1. Convert the image into the frequency domain using FFT.
2. Design a Butterworth Low-Pass Filter with an appropriate cutoff frequency.
3. Apply the filter to suppress high-frequency noise.
4. Perform the inverse FFT to reconstruct the image.

How It Achieves the Desired Effect

- Preserves low-frequency information.
- Gradually attenuates high-frequency components.
- Produces smoother images with minimal distortion.
- Avoids sudden transitions found in Ideal LPF.

Advantages

- Smooth frequency response.
- Better edge preservation than Ideal LPF.
- Reduces ringing artifacts.
- Effective for moderate noise removal.

Applications

- Medical imaging.
- Face recognition.
- Remote sensing.
- Industrial inspection.

Conclusion

The Butterworth Low-Pass Filter effectively preserves important image information while reducing unwanted noise, making it suitable for image enhancement.

3. An image requires edge enhancement in the frequency domain. Explain and justify the filtering approach used, describing its effect on frequency components.

A.

Introduction

Edges correspond to rapid intensity changes and are represented by high-frequency components in an image. To enhance edges, a High-Pass Filter (HPF) is applied.

Suitable Filter

Gaussian High-Pass Filter (GHPF) or Butterworth High-Pass Filter (BHPF).

Working

1. Convert the image into the frequency domain using FFT.
2. Apply the High-Pass Filter.
3. Low-frequency components are suppressed.
4. High-frequency components are preserved and enhanced.
5. Apply the inverse FFT to obtain the enhanced image.

Effect on Frequency Components

- Removes low-frequency background information.
- Enhances high-frequency edge details.
- Makes object boundaries sharper.
- Improves feature visibility.

Advantages

- Enhances image sharpness.
- Highlights edges and textures.
- Useful for feature extraction.
- Improves object recognition accuracy.

Applications

- Medical diagnosis.
- Fingerprint recognition.
- Satellite image analysis.
- Industrial quality inspection.

Conclusion

A High-Pass Filter is the best choice for edge enhancement because it emphasizes high-frequency information while suppressing smooth background regions, resulting in clearer and sharper edges.

4. A smooth version of an image is required. Describe and justify the frequency domain technique used for smoothing, explaining its operation and results.

A.

Introduction

Image smoothing is used to reduce noise and minor intensity variations while preserving the overall appearance of an image. In the frequency domain, smoothing is achieved using a Low-Pass Filter (LPF).

Suitable Technique

Gaussian Low-Pass Filter (GLPF)

Working

1. Convert the image into the frequency domain using the Fourier Transform (FFT).
2. Design a Gaussian Low-Pass Filter with an appropriate cutoff frequency.
3. Multiply the filter with the image spectrum.
4. Apply the Inverse Fourier Transform (IFFT) to reconstruct the smoothed image.

Results

- Reduces random noise.
- Produces a smooth image.
- Preserves important structures.
- Minimizes ringing effects.

Advantages

- Excellent smoothing performance.
- Reduces image noise.
- Produces natural-looking images.
- Computationally efficient.

Applications

- Medical image preprocessing.
- Satellite image enhancement.
- Face recognition.
- Noise reduction before segmentation.

Conclusion

The Gaussian Low-Pass Filter effectively smooths an image while maintaining its important visual information.

5. An object is clearly brighter than the background. Explain and justify the segmentation technique used, including its working and effectiveness.

A.

Introduction

Segmentation divides an image into meaningful regions. When the object is much brighter than the background, Threshold Segmentation is the most suitable technique.

Suitable Technique

Global Thresholding

Working

1. Select a threshold value (T).
2. Compare every pixel intensity with T.
3. Pixels greater than T become the object.

4. Pixels less than T become the background.

Mathematical Expression

If $f(x,y) > T$, Object (White)

Else, Background (Black)

Effectiveness

- Simple implementation.
- Fast computation.
- Works well for high-contrast images.
- Produces accurate object extraction.

Advantages

- Easy to implement.
- High processing speed.
- Effective for simple images.
- Low computational complexity.

Applications

- Document image processing.
- Cell detection.
- Industrial inspection.
- OCR systems.

Conclusion

Threshold segmentation is highly effective when there is a clear intensity difference between the object and the background.

6. You need to separate foreground and background based on intensity. Describe and apply an appropriate segmentation method, providing a clear explanation of its process.

A.

Introduction

Separating foreground from background is a common image segmentation task. Otsu's Thresholding is widely used because it automatically determines the optimal threshold.

Suitable Method

Otsu Thresholding

Working

1. Compute the image histogram.
2. Calculate the variance for every possible threshold.
3. Select the threshold that maximizes between-class variance.
4. Segment the image into foreground and background.

Process

- Analyse histogram.
- Determine optimal threshold automatically.
- Convert image into binary form.
- Separate foreground and background.

Advantages

- Automatic threshold selection.
- High segmentation accuracy.
- No manual parameter tuning.

- Simple implementation.

Applications

- Medical imaging.
- Document processing.
- Machine vision.
- Object detection.

Conclusion

Otsu's Thresholding provides reliable foreground-background separation without manual threshold selection.

7. An image requires identification of object edges. Explain and justify a suitable edge detection method, describing its working and effectiveness.

A.

Introduction

Edge detection identifies boundaries between objects by detecting sudden intensity changes.

Suitable Method

Canny Edge Detection

Working

1. Smooth image using Gaussian Filter.
2. Calculate image gradients.
3. Apply Non-Maximum Suppression.
4. Perform Double Thresholding.
5. Use Edge Tracking by Hysteresis.

Effectiveness

- Detects true edges accurately.
- Reduces noise influence.
- Produces thin edges.
- Minimizes false detections.

Advantages

- High accuracy.
- Good localization.
- Noise resistant.
- Produces continuous edges.

Applications

- Face detection.
- Medical image analysis.
- Autonomous vehicles.
- Industrial inspection.

Conclusion

Canny Edge Detection is considered one of the best edge detection algorithms due to its accuracy and robustness.

8. Adjacent pixels have similar intensity values. Describe and apply a segmentation technique to group such pixels, explaining the concept and reasoning.

A.

Introduction

When neighboring pixels have similar intensity values, Region Growing Segmentation is commonly used.

Suitable Technique

Region Growing

Working

1. Select seed pixels.
2. Compare neighboring pixel intensities.
3. Merge similar neighboring pixels.
4. Continue until no similar pixels remain.

Concept

Pixels with similar intensity and connectivity belong to the same region.

Advantages

- Produces connected regions.
- High segmentation accuracy.
- Simple implementation.
- Suitable for homogeneous images.

Applications

- Medical image segmentation.
- Satellite image analysis.
- Object extraction.
- Tumor detection.

Conclusion

Region Growing effectively groups similar neighboring pixels into meaningful regions.

9. Detected edges in an image appear broken.

Explain and justify a method to improve edge continuity, including its role in post-processing.

A.

Introduction

Broken edges reduce the quality of edge detection and affect object recognition. Morphological operations are commonly used to improve edge continuity.

Suitable Method

Morphological Closing

Working

1. Apply Dilation.
2. Apply Erosion.
3. Fill small gaps.
4. Connect broken edges.

Role in Post-processing

- Connects nearby edge segments.
- Removes small holes.
- Improves object boundaries.

- Produces continuous edges.

Advantages

- Enhances edge continuity.
- Removes small gaps.
- Improves segmentation quality.
- Easy implementation.

Applications

- OCR.
- Medical imaging.
- Object tracking.
- Pattern recognition.

Conclusion

Morphological Closing effectively improves edge continuity and enhances overall segmentation quality.

10. You need to extract object outlines from an image. Describe and apply an appropriate technique, explaining how it identifies boundaries effectively

A.Introduction

Object outline extraction identifies the boundary separating an object from the background.

Suitable Technique

Contour Detection

Working

1. Convert image to grayscale.
2. Apply thresholding.
3. Detect edges.
4. Find contours.
5. Draw contours around detected objects.

Boundary Identification

- Detects connected edge pixels.
- Traces object boundaries.
- Represents object shape.
- Extracts complete outlines.

Advantages

- **Accurate boundary extraction.**
- **Shape analysis.**
- **Efficient implementation.**
- **Useful for object recognition.**

Applications

- **Shape recognition.**
- **Medical image analysis.**
- **Industrial inspection.**
- **Computer vision systems.**

Conclusion

Contour Detection accurately extracts object outlines and provides valuable information for object analysis and recognition.